

Auteur: Anna Voronovska
Studierichting: Informatica
Oefeningenreeks 3D nr. 3

$$f(x) = ax^3 + bx^2$$

Stap 1

$$f'(x) = 3ax^2 + 2bx$$

$$f'(2) = 3a \cdot 2^2 + 2b \cdot 2 = 12a + 4b$$

$$f'(x) = 0, \text{ dus:}$$

$$12a + 4b = 0$$

$$3a + b = 0 \text{ (*)}$$

Stap 2

$$f(2) = a \cdot 2^3 + b \cdot 2^2 = 8a + 4b$$

$$f(2) = -4$$

Dus:

$$8a + 4b = -4$$

$$2a + b = -1 \text{ (**)}$$

Stap 3: Stelsel:

$$\begin{cases} 3a + b &= 0 \\ 2a + b &= -1 \end{cases}$$

$$\begin{cases} b &= -3a \\ 2a - 3a &= -1 \end{cases}$$

$$\begin{cases} b &= -3a \\ -a &= -1 \end{cases}$$

$$\begin{cases} b &= -3 \cdot 1 \\ a &= 1 \end{cases}$$

$$\begin{cases} b &= -3 \\ a &= 1 \end{cases}$$

Antwoord: $a = 1$; $b = -3$.

References